

Kindle Vocabulary Plugin for Obsidian

Submitted as the final report for the

Course-related Project II

by

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ABSTRACT

The Obsidian is a novel note taking system, which supports the user to store all of their notes in the local machine instead of in the cloud as other services. The unique point that makes Obsidian stand out is the huge community of developers who daily contribute to the plugins store, which provides the flexibility for the user to customize their note-taking system. Meanwhile, Kindle is a reading machine created by Amazon, one of the main features of the Kindle is the Vocabulary Builder, which enables user to learn the words that they have searched, including the context, and the book, in which the words have been looked up. This project aim to develop a plugin that helps the user export/organize all of the information in the Kindle Vocabulary Builder to their Obsidian.

Keywords: Obsidian, Obsidian's plugin, Kindle Vocabulary Builder.

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Chapter 1

INTRODUCTION

The project emerged from a personal pain point - the lack of a simple way to export Kindle Vocabulary Builder data into Markdown for use in Obsidian.

1.1 Project Background

- The “Kindle Highlights for Obsidian” plugin by Hadynz ([GitHub](#)) demonstrates extracting Kindle highlight data (stored as text in `My Clippings.txt`) and syncing it to Obsidian, providing a structural reference for file extraction and plugin design.
- The “Kindle Vocabulary Builder Export” tool by Henry Wang ([henrywang.nl](#)) outlines methods to access Vocabulary Builder data, stored in non-text database files (e.g., `vocab.db`), but lacks integration with note-taking platforms. Moreover, it needs to be noted that this implementation requires the user to DeDRM the dictionary, which is illegal in some regions.
- Obsidian’s plugin development documentation ([Obsidian Developer Docs](#)) details plugin anatomy, including TypeScript/JavaScript frameworks, essential for building custom functionality.

1.2 Project Definition

1.2.1 Problem Statement

In this project, we aim to build a simple way for the Obsidian user to export their Kindle’s Vocabulary Builder to their Markdown Note in Obsidian.

1.2.2 Context and Scope

We mainly implement these final features for the user in our plugin:

- Generate Markdown from your Kindle lookups (vocab.db)
- Integrate custom dictionary files (CSV format)
- Mark words as "Learned/Unlearned" using checkboxes
- View statistics such as percentage learned
- Sort words by timestamp or prioritize unlearned items

1.2.3 Significance and Implications

There was a plugin, that helps the user to export their Kindle's highlight note to Obsidian, and that plugin got over a thousand stars in [GitHub](#), which indicates that the number of users who use both Obsidian and Kindle are huge. And it was quite surprised that there is no plugin that helps user to export their Vocabulary Builder, which is one of the main features of Kindle, besides the highlight note.

1.2.4 Quantification (if any)

Chapter 2

PROJECT MANAGEMENT

2.1 Project Plan

Week	Date Range	Tasks
1	Apr 14–20	Study Obsidian plugin anatomy, analyze “Kindle Highlights” repository, set up GitHub repository.
2	Apr 21–27	Develop basic plugin for random Kindle file extraction, test in Obsidian. <i>Deliverable: Phase 1 repository.</i>
3	Apr 28–May 4	Investigate Kindle folder structure, parse vocab.db with SQLite3 (midterm preparation).
4	May 5–11	Finalize vocab.db extraction, convert extracted data to mark-down format.
5	May 12–18	Integrate extraction functionality into plugin, test import process. <i>Deliverable: Phase 2 repository.</i>
6	May 19–25	Refine plugin UI, clean codebase, draft project report, prepare demo video. Research and implement IPA pronunciation support. Design a system to mark words as “learned” or “unlearned”.
7	May 26–Jun 1	Add MIT license, finalize documentation, publish plugin, submit deliverables. Write blog post to introduce plugin on Obsidian’s forum. Explore integration with Anki for vocabulary syncing, inspired by existing repositories.
8	Jun 2–8	Investigate feasibility of direct sync with Amazon accounts to eliminate manual vocab.db uploads.
9	Jun 9–15	Finalize Amazon sync research, implement if feasible, or document findings. Perform final testing and bug fixes.
10	Jun 16–20	Finalize all project components, ensure all deliverables are submitted, and publish final blog post updates.

2.2 Contribution of Team Members

Both team members, Truong Gia Bao and Nguyen Canh Huy, contributed equally to all aspects of the project, ensuring a balanced and collaborative effort throughout the development process from April 14 to June 20, 2025. Their contributions encompassed the following:

- **Plugin Development:** Both members actively participated in studying the Obsidian plugin anatomy, analyzing the “Kindle Highlights” repository, developing the plugin for Kindle file and vocab.db extraction, implementing IPA pronunciation support, designing the learned/unlearned system, exploring Anki integration, and researching Amazon sync feasibility.
- **Testing and Refinement:** Both collaborated on testing the plugin in Obsidian, refining the user interface, cleaning the codebase, and ensuring robust functionality across all features.
- **Deliverables:** Both contributed to creating and submitting the Phase 1 and Phase 2 repositories, preparing the demo video, drafting and finalizing the project report, and writing the blog post for Obsidian’s forum.
- **Documentation and Licensing:** Both worked on adding the MIT license, finalizing comprehensive documentation, and ensuring all deliverables met project requirements.
- **Presentation:** Both team members jointly prepared and reviewed presentation slides to effectively communicate the project’s objectives, progress, and outcomes.

The equal division of responsibilities ensured that all tasks, from technical development to documentation and presentation, were completed collaboratively, leveraging the strengths of both team members to deliver a high-quality Obsidian plugin.

2.3 Challenges and Decision Making

Challenge 1: DRM-Protected Kindle Dictionary

The initial plan was to utilize the dictionary provided by Kindle to export vocabulary information for the plugin. However, this dictionary was **DRM-protected**, and using

DeDRM tools to access it was deemed illegal due to copyright restrictions. This posed a significant barrier to extracting comprehensive word data, including definitions and pronunciations, necessary for the plugin's functionality.

Decision: To adhere to legal and ethical standards, the team opted to use an open-source alternative, Webster's Dictionary, available at <https://www.gutenberg.org/ebooks/29765>. Both team members evaluated this dictionary as a viable substitute. However, it presented limitations: insufficient word coverage, lack of International Phonetic Alphabet (IPA) transcriptions, and challenges in formatting derived word forms. To overcome these shortcomings, the team explored Wiktionary, which offered a more extensive and structured dataset. A Python script was developed to call the Wiktionary API, programmatically extracting word data to create a custom dictionary tailored to the plugin's needs. To provide flexibility, both the Webster's Dictionary and the Wiktionary-derived dictionary were included in the GitHub repository, allowing users to choose based on their preferences.

Challenge 2: Direct Amazon Account Sync

The team aimed to implement direct synchronization with Amazon accounts to eliminate the need for users to manually upload the vocab.db file, enhancing user experience and automation. However, after thorough investigation, it was determined that Amazon does not provide a suitable API to support direct access to Kindle vocabulary data, making this feature infeasible within the project timeline.

Decision: The team decided to retain the manual vocab.db upload process, ensuring the plugin remained functional and user-friendly despite the lack of direct sync. Both team members documented the manual process clearly in the plugin's documentation and explored potential workarounds, such as parsing local Kindle folder structures, to streamline data access. The findings from the Amazon sync research were included in the project report, and the team noted that future iterations could revisit this feature if Amazon releases a suitable API.

Chapter 3

SYSTEM DESCRIPTION

3.1 Key architecture of the system

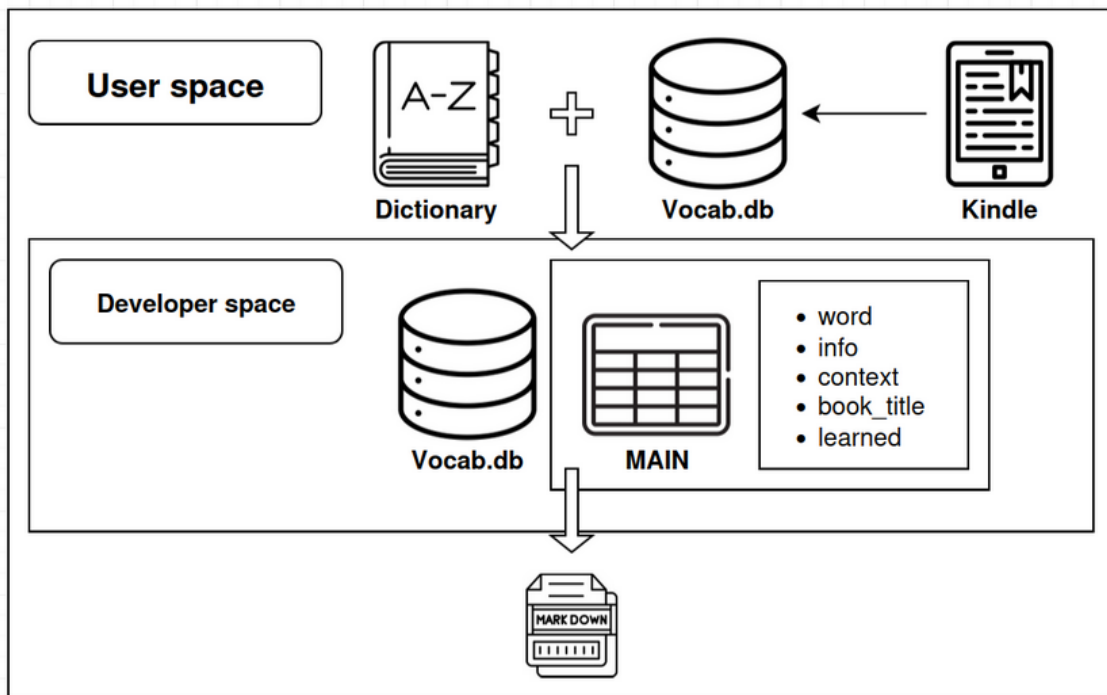


Figure 3.1: Key architecture of the system

3.2 Architecture Explanation

The architecture of the Obsidian plugin project, developed between April 14 and June 20, 2025, is designed to facilitate vocabulary management by integrating data from a

Kindle device with a dictionary, processing it into a structured database, and exporting it in markdown format for use within Obsidian. The system is divided into two primary spaces—User space and Developer space—each playing a critical role in the workflow, as illustrated in the architectural diagram.

3.2.1 Data Acquisition

In the User space, the process begins with the retrieval of the `vocab.db` file, which is obtained by directly connecting the Kindle device to a laptop or PC. This file contains raw vocabulary data logged by the Kindle, including words looked up by the user. Additionally, a dictionary (either Webster’s or the Wiktionary-derived dictionary, as provided in the GitHub repository) is supplied to enrich the vocabulary data with definitions and other linguistic information. These two data sources—`vocab.db` and the dictionary—are combined to form the foundation of the system’s data processing.

3.2.2 Data Processing in Developer Space

Within the Developer space, the `vocab.db` file is further processed to create a centralized database structure. The core component is the `MAIN` table within `vocab.db`, which integrates data from the Kindle and the dictionary. This table is designed with the following columns:

- `word`: The vocabulary word looked up by the user.
- `info`: The definition or additional information sourced from the dictionary.
- `context`: The context in which the word was looked up, extracted from the Kindle data.
- `book-title`: The title of the book where the word was encountered, retrieved from the Kindle metadata.
- `learned`: A boolean field to track whether the word has been marked as learned or unlearned by the user.

This integration is performed programmatically, ensuring that each word entry is enriched with contextual and definitional data, enabling effective vocabulary management within the plugin.

3.2.3 Output Generation

A custom Python script, developed as part of the project, is employed to export the data from the MAIN table into markdown format. This script queries the database, formats the information into a structured markdown file, and outputs it for use in Obsidian. The markdown files include the word, its definition, context, book title, and learned status, allowing users to review and manage their vocabulary seamlessly within the Obsidian environment. The exported markdown files are generated as part of the plugin's functionality, providing a user-friendly interface for tracking progress and integrating with other Obsidian features.

3.3 The implementation structure

The key logics, which have already been described in the previous part, lies mainly in the modals folder, and the export from the MAIN table to Obsidian note is in the MarkdownFormat.ts file. For more information about the Ribbon, Settings, and the Obsidian's plugin anatomy, please refer to this [link](#).

Directory/File	Description
src/	Root directory for source code
commands/	Directory for command-related files
CommandsHandlers.ts	Registers and defines custom plugin commands
data/	(Empty or reserved) For future data storage or constants
modals/	Directory for modal-related files
DatabaseUploadModal.ts	Modal to upload Kindle's vocab.db file
DictionaryUploadModal.ts	Modal to upload custom dictionary CSV
SyncDatabaseModal.ts	Modal for syncing database and generating Markdown
ribbon/	Directory for ribbon-related files
RibbonHandlers.ts	Adds and handles ribbon icon behavior
settings/	Directory for settings-related files
Settings.ts	Defines the plugin's settings schema and defaults
SettingTab.ts	Renders the settings tab UI in Obsidian
utils/	Directory for utility files
MarkdownFormat.ts	Generates interactive Markdown output and listeners
PathHelper.ts	Path resolution for vault, plugin data, etc.
main.ts	Entry point: initializes plugin, settings, and features

Chapter 4

SYSTEM TESTING AND ANALYSIS

We perform the manual testing to check all of our features, including:

- Generate Markdown from your Kindle lookups (`vocab.db`)
- Integrate custom dictionary files (CSV format)
- Mark words as "Learned/Unlearned" using checkboxes
- View statistics such as percentage learned
- Sort words by timestamp or prioritize unlearned items

4.1 Generate the Markdown file, and Integrate custom dictionary files

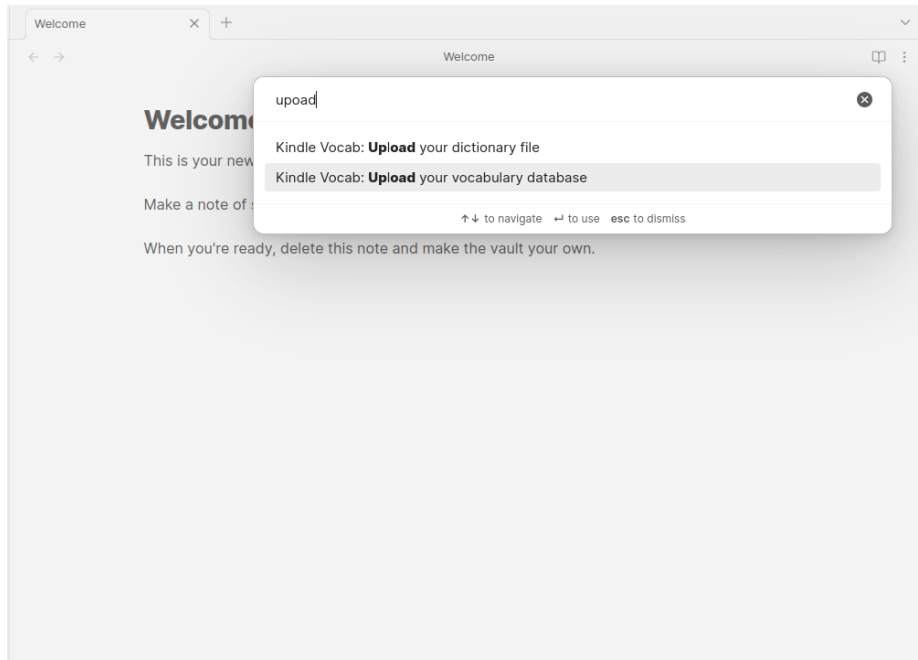


Figure 4.1: The UI of upload vocab.db and dictionary.csv

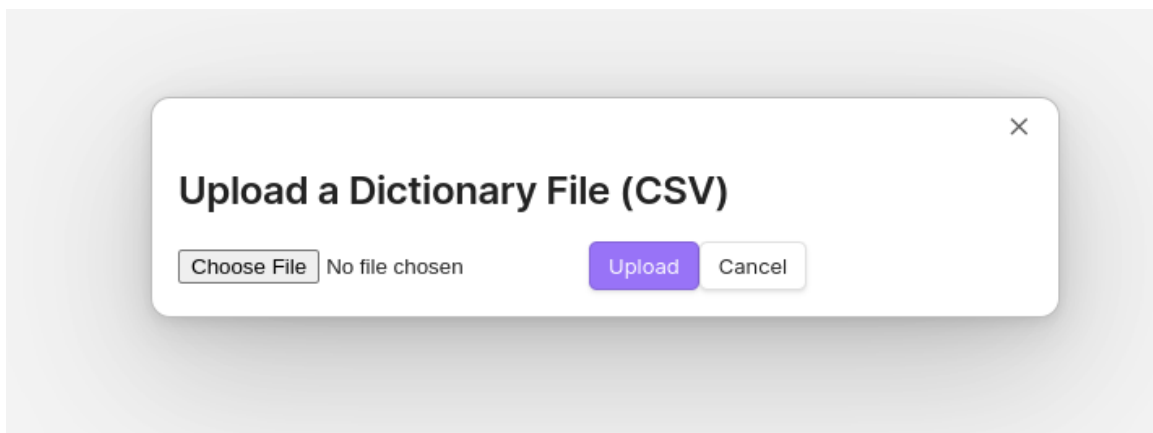


Figure 4.2: Upload button

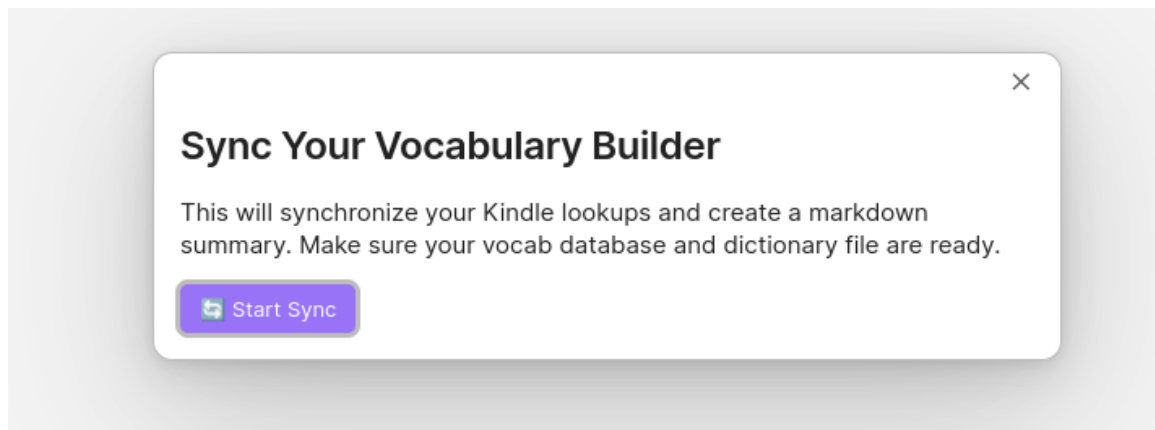


Figure 4.3: Sync button

My Vocabulary Builder

Recent Lookups

You've learned 0 out of 516 words (0%)

lodging

- **Context:** This lodging of yours I had forgotten, though I never remembered it, indeed, because I did not know it; and as for your old lodgings, I could only remember it was at the Five Corners, Harlamov's house.
- **Book Title:** Crime and Punishment: The Original Unabridged And Complete Edition (A Fyodor Dostoevsky Classics)
- **Learned:** ☐
- **Definition:**

/lɒdʒɪŋ/
Noun

1. A place to live or lodge.
2. Sleeping accommodation.
3. Furnished rooms in a house rented as accommodation.
- 4.
5. The condition of a plant, especially a cereal, that has been flattened in the field or damaged so that it cannot stand upright, as by weather conditions or because the stem is not strong enough to support the plant.

Verb

1. present participle and gerund of lodge

raspberries

- **Context:** "And where is she to get raspberries for you?"
- **Book Title:** Crime and Punishment: The Original Unabridged And Complete Edition (A Fyodor Dostoevsky Classics)



Figure 4.4: The result

4.2 Mark words as "Learned/Unlearned"

To use this feature, the user need to switch to Preview mode, by using **Ctrl + E**.

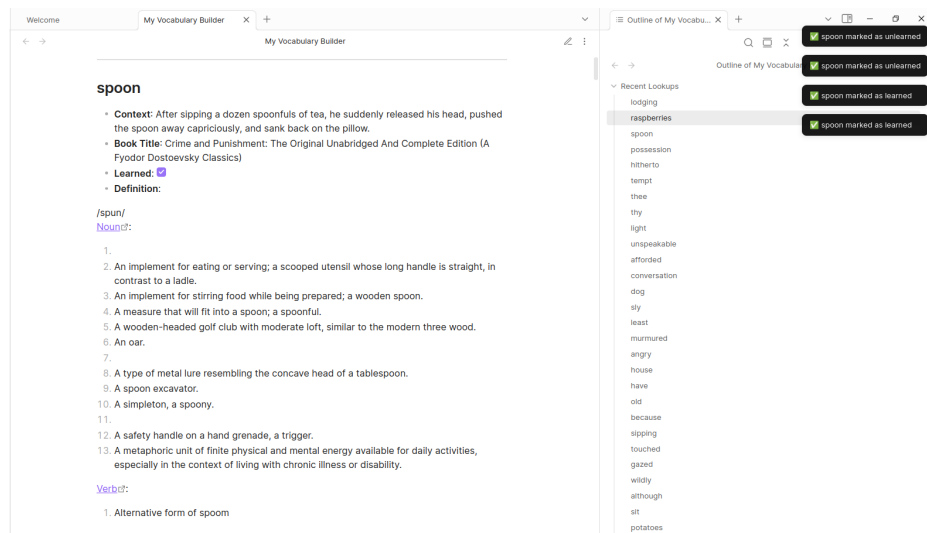


Figure 4.5: Learned/unlearned feature

After click to learned box, and resync the file, we get the following result:

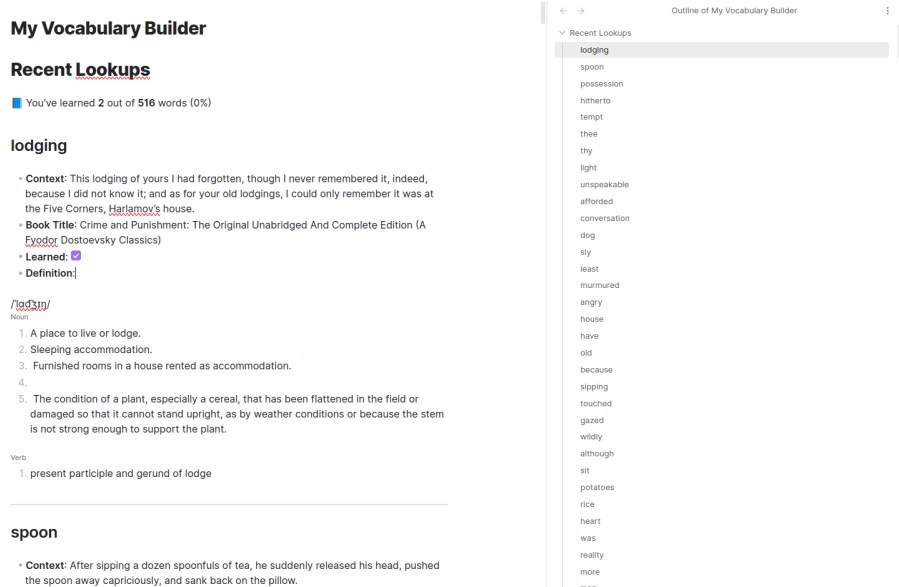


Figure 4.6: After resyncing

4.3 Sort options

By accessing to the settings and changing the sort options, we can choose which sort order we want the system to perform.

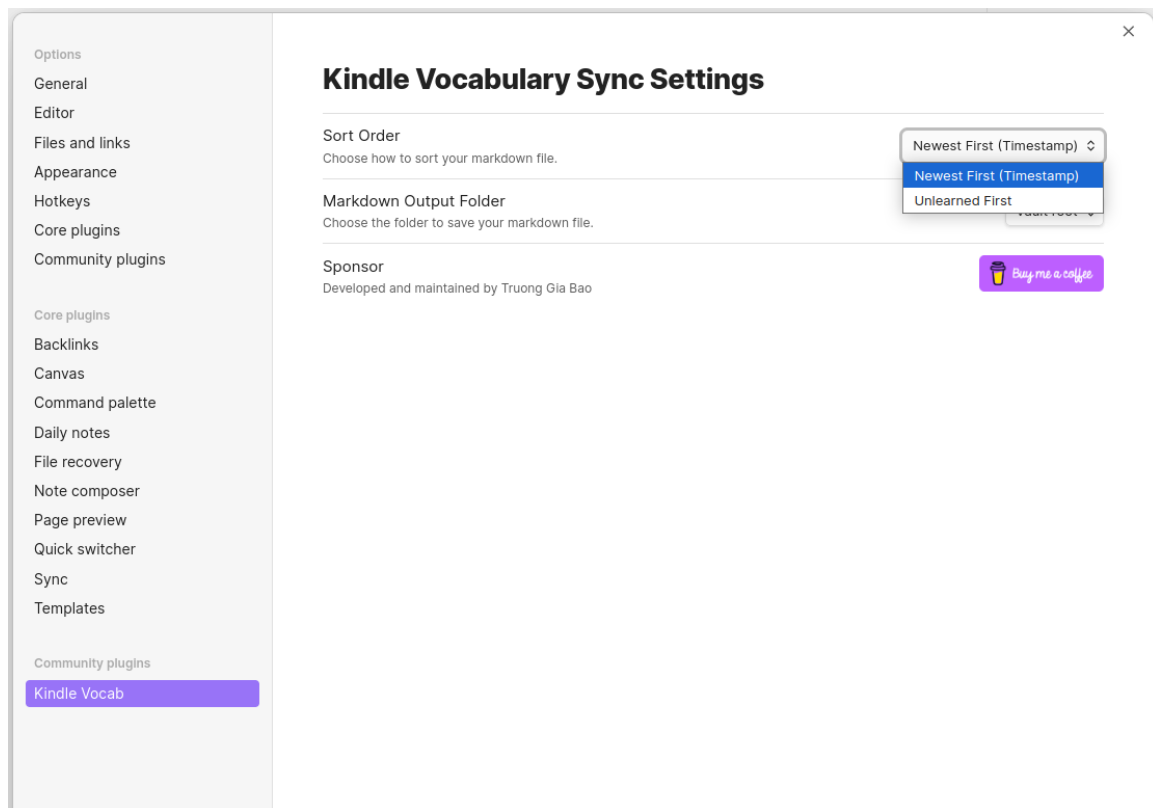


Figure 4.7: Settings pannel

We can see that after we choose the "Unlearned" first, the learned words in previous part, "lodging", and "spoon" are placed at the end of the note.

My Vocabulary Builder

Recent Lookups

You've learned 2 out of 516 words (0%)

possession

• **Context:** But from some queer, almost animal, cunning he conceived the idea of hiding his strength and lying low for a time, pretending if necessary not to be yet in full possession of his faculties, and meanwhile listening to find out what was going on.

• **Book Title:** Crime and Punishment: The Original Unabridged And Complete Edition (A Fyodor Dostoevsky Classics)

• **Learned:** ☐

• **Definition:**

Noun

1. Control or occupancy of something for which one does not necessarily have private property rights.
2. Something that is owned.
3. Ownership; taking, holding, keeping something as one's own.
4. A territory under the rule of another country.
- 5.
6. The condition or affliction of being possessed by a demon or other supernatural entity.
- 7.
8. The condition of being under the control of strong emotion or madness.
9. Control of the ball; the opportunity to be on the offensive.
10. A disposal of the ball during a game, i.e. a kick or a handball.
11. A syntactic relationship between two nouns or nominals that may be used to indicate ownership.

Verb

1. To invest with property.

← →

Outline of My Vocabulary Builder

Individuals
arteries
coronary
classified
Inadequate
doubling
demolishes
enthralled
dwarf
Adequate
Plentiful
yours
prescription
function
stubborn
Drowsy
coincidence
aforementioned
eroded
fabric
societal
wellness
component
legitimate
deprive
futile
attempt
disruption
determining
Insufficient
I
lodging
spoon

Figure 4.8: After editing the settings

Chapter 5

CONCLUSION AND DISCUSSION

5.1 Conclusion

- We have successfully created the Plugin for Obsidian to help the user export their vocabulary builder in Kindle to their note taking system. With many additional features, including, mark words as learned/unlearned, and allow the user to sort words based on that, track the percentage of learning words.
- Currently, we submitted the [Pull Request](#) to the Obsidian team to review it. We believe that after getting the official publish, we can get at least 100 stars for our [repository](#) in Github.
- We also wrote the [Developer's documentation](#) for those who want to explore and contribute to our project.

5.2 Future Developments

- Handle the dictionary for other languages, not only English.
- Handle the deviating word forms better, for example, if the user search "raspberries" instead of "raspberry", the result will not show up.
- Enhance the current dictionary, there are still many uncommon words that are not in the dictionary.
- The current developer's documentation doesn't contain the system architect and how it work, we will fix it as soon as possible. ?

REFERENCES

- [Obsidian Plugin Developer Docs](#) - Official documentation for building plugins in Obsidian.
- [Obsidian API Reference](#) - API reference for Obsidian plugin development.
- [TypeScript Handbook](#) - Comprehensive guide to the TypeScript programming language.
- [esbuild Documentation](#) - Documentation for the esbuild JavaScript bundler.
- [GCIDE Dictionary](#) - GNU Collaborative International Dictionary of English.
- [Webster's Dictionary CSV](#) - Open-source CSV version of Webster's English Dictionary.
- [Wiktionary](#) - Collaborative online dictionary and thesaurus.
- [Microsoft Docs – DRM](#) - Microsoft documentation on Digital Rights Management.

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